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Tablet-based interventions for individuals with ASD: Evidence of generalization and maintenance effects

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ABSTRACT

Background: Despite positive effects of and established empirical evidence for tablet-based interventions for individuals with autism spectrum disorder (ASD), it is not known whether such findings can be applicable to maintenance and generalization effects of those interventions.

Aims: This systematic review evaluated peer-reviewed single-case experimental design (SCED) studies concerning evidence of generalization and maintenance effects of tablet-based interventions for individuals with ASD.

Methods: To evaluate the methodological rigor of the literature in terms of generalization and maintenance effects, the authors adopted four of the basic design standards developed by What Works Clearinghouse (WWC; Kratochwill et al., 2010/2014). In addition, the authors used Tau-U effect size measure and attempted to calculate effect sizes differentiated by the type of generalization and maintenance teaching strategies.

Results: A total of 21 studies assessed generalization and/or maintenance effects. In the first evaluation of evidence of generalization and maintenance effects, it was found that more than half of the studies included in this review collected interobserver agreement (IOA) on at least 20% of sessions across all generalization and maintenance conditions and met the minimum quality thresholds of IOA. Meanwhile, less than one third of the studies included more than three data points in each generalization and maintenance condition. With regard to maintenance of effects, about half of the reviewed studies did not report the latency to the maintenance measure, which may hamper the assessment of the clinical and practical significance of the effect of the tablet-based intervention. In the second evaluation, the omnibus Tau-U effect size for baseline to generalization comparisons resulted in a moderate effect. For the contrasts between intervention and generalization comparisons resulted in a small effect. The omnibus Tau-U effect size for baseline and maintenance comparisons resulted in a strong effect. For the contrasts between intervention to maintenance comparisons, the omnibus Tau-U effect size resulted in a small effect.

Conclusions and implication: Findings in this review suggest that efforts should be made to establish a system for appraising generalization and maintenance procedures in SCED studies. In

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addition, future studies should investigate if tablet-based interventions are truly effective in creating sustainable behavioral change in individuals with ASD.

What this paper adds?

Given the proliferation of mobile technology, many researchers and practitioners adopt tablet-based interventions when educating individuals with ASD. Accordingly, many SCED studies have established empirical evidence of tablet-based interventions on learning in those individuals. However, it is not clear whether such findings would be consistent when considering evidence of generalization and maintenance effects of the interventions. In this study, we evaluated the literature on tablet-based interventions for individuals with ASD against the design standards to identify evidence of generalization and maintenance effects of the interventions. In addition, we calculated effect sizes differentiated by the type of generalization and maintenance teaching strategies. As a result, we found that more than half of the studies included met the design standards concerning IOA measures while only a few studies met the standard concerning the number of data points per phase. Further, we found that about half of the articles in the review employed a programming common stimuli as a generalization and/or maintenance teaching strategy.

1. Introduction

1.1. Tablet-based interventions and ASD

The number of people diagnosed with autism spectrum disorder (ASD) has exploded in recent years, with the current prevalence rate estimated to be 1 in 68 individuals (Centers for Disease Control and Prevention, 2014). People diagnosed with ASD often have deficits in social communication and interaction and display restrictive or repetitive behaviors, activities, or interests (American Psychiatric Association, 2013). In addition to these characteristics, people with ASD may also have difficulty generalizing and maintaining learned skills when they are not specifically programmed for by interventionists (Brown & Bebko, 2012; Foxx, 2008). Maintenance, also termed generalization across time, is especially important to program for because failure to do so may necessitate the indefinite delivery of an intervention at full strength (Freeland & Noell, 2002). Additionally, failure to program for generalization across people, settings, or behaviors can result in the acquisition of skills that individuals are only able to demonstrate in contrived settings or with a limited number of people (Neely et al., 2016). Because the goal of most interventions is lasting behavior change in multiple contexts, it is crucial for interventionists to purposefully plan for generalization and maintenance before an intervention is initiated (Freeland & Noell, 2002).

One intervention that has been gaining popularity for children with ASD is tablet-based interventions. Tablet-based interventions include any intervention that uses a tablet-based computer (e.g., iPad, iPod, Android device, etc.) for the purpose of increasing an individual's social, academic, behavioral, or communication skills. The use of tablet-based computers has greatly increased over the years (Common Sense Media, 2013), and many parents and teachers are using tablet-based interventions in an attempt to remediate some of the deficits that are inherent in ASD (Clark, Austin, & Craike, 2015; King, Thomeczek, Voreis, & Scott, 2014). While the results of two recent meta-analyese indicate that tablet-based interventions are effective (Hong et al., 2017; Stephenson & Limbrick, 2015), both studies limited their statistical analyses to comparisons between baseline and intervention phases; thus, it is still unknown whether the positive results found for tablet-based interventions maintained in the absence of intervention and generalized to other people, settings, behaviors, and materials.

1.2. Evidence of generalization and maintenance effects and methodological rigor

Over the last few years, great emphasis has been placed on identification of educational practices that have empirical evidence of effectiveness for individuals with ASD. When determining intervention effectiveness, it must be questioned if these effects will maintain in the absence of intervention as well as whether or not findings can be generalized outside the experimental conditions (i.e., external validity; Johnson & Christensen, 2000). These questions can be answered by the inclusion of generalization and maintenance data; however, many SCED studies still do not report generalization and maintenance effects (see Neely, Davis, Davis, & Rispoli, 2015), despite recommendations in the field to include these measures (Neely et al., 2016). This is concerning, considering failure to include generalization and maintenance may lower the methodological rigor of those studies, and limit the effects of the intervention to simulated settings (Neely et al., 2016).

When conducting a systematic review of the literature, it is important to document whether studies included generalization and maintenance measures and to evaluate the quality of these measures when they are included; however, as yet, there are no guidelines for evaluation generalization and maintenance effects and the validity of evidence. While several approaches have been proposed to evaluate generality of treatment effects that enhances external validity (e.g., Horner et al., 2005; Maggin, Briesch, & Chafouleas, 2013; Reichow, Volkmar, & Cicchetti, 2008; Wolery, 2013), one commonality among these approaches is the important of operational descriptions of participants, settings, and dependent and independent variables; however, none of these clearly refines specific thresholds for determining whether or not given descriptions on the study would promote replication of the effects beyond those defined in the study (Hitchcock, Kratochwill, & Chezan, 2015). Furthermore, little or nothing is known about what criteria should be

used to evaluate evidence of generalization and maintenance effects regarding external validity in SCED studies (Hitchcock et al., 2015). The development of these guidelines would help researchers better understand the state of evidence regarding generalization and maintenance effects in SCED studies.

In addition to evaluating SCED research in terms of external validity, it is also important to evaluate the internal validity of studies by using a set of indicators to determine research quality and methodological soundness. Currently, several attempts have been made to develop rubrics that can be used to evaluate and quantify the methodological rigor in SCEDs (e.g., Cook et al., 2014; Horner et al., 2005; Kratochwill et al., 2010; Kratochwill, 2013; Maggin et al., 2013; Odom et al., 2005). These rubrics contain multiple elements that evaluate a study's internal validity, including procedural fidelity, inter-observer agreement, the systematic manipulation of the independent variable, and replication of treatment effects. Although these existing rubrics have helped researchers identify practices that have strong evidence for individuals with ASD, there seems little agreement on what thresholds should be used to evaluate evidence of generalization and maintenance effects (Hitchcock et al., 2015); however, given a well-defined set of thresholds in the existing rubrics, it seems relatively straightforward to quantify.

1.3. Generalization and maintenance effects and effect sizes

When evaluating a body of literature, it is important to determine if studies assess generalization and maintenance of learned skills. For maintenance in particular, it is important to determine how long after the conclusion of intervention is assessed. Recent reviews of interventions for individuals with ASD have reported that maintenance is not often assessed or planned for in the literature, and when it is planned, it is typically assessed soon after the conclusion of intervention (McDonald & Machalicek, 2013; Neely et al., 2016; Palmen, Didden, & Lang, 2012). Reports of the frequency for which generalization is assessed for this population are also dismal (McDonald & Machalicek, 2013; Neely et al., 2016) and indicate that studies are not actively planning for generalization. These results are discouraging considering the goal of most interventions is lasting behavior change that extends beyond the conditions and settings in the study. Determining whether the literature base on tablet-based intervention studies for individuals with ASD also fails to assess generalization and maintenance, and the latency to maintenance when it is assessed, provides valuable information to researchers and practitioners about the state of the literature on this topic.

In addition to determining whether generalization and maintenance are assessed, it is also important to determine what strategies studies typically use to program for generalization. In their seminar article on generalization, Stokes and Baer (1977) reviewed hundreds of articles that programmed for generalization, and from their review, they created nine categories to describe the techniques that studies used to program for generalization. These categories include train and hope, sequential modification, introduction to natural maintaining contingencies, training sufficient exemplars, training loosely, using indiscriminable contingencies, programming for common stimuli, mediating generalization, and training to generalize. Each of these categories, except for the train and hope category, requires the practitioner or researcher to actively program for generalization. Stokes and Baer (1977) described train and hope as a first step in an analysis of generalization, with the intention that generalization would be programmed for by including some of the other categories if generalization did not spontaneously occur; however, many researchers use train and hope as their sole technique for generalization and do not actively program for it if it does not occur naturally (Neely et al., 2016; Schlosser & Lee, 2000). Using the categories proposed by Stokes and Baer (1977) to document the strategies studies use when they do program for generalization reveals gaps in the literature and provides practitioners with information that they can use to determine if a particular intervention has sufficient evidence of generalization to warrant its use with a client. Furthermore, when reviewing a body of literature for generalization and maintenance measures, it is also important to calculate an effect size to determine if the targeted intervention has differential effects on categories of variables, such as the generalization and maintenance teaching strategies. By using an effect size to calculate the effects of tablet-based interventions on the generalization and maintenance of different teaching strategies, practitioners are provided with information about for whom and under what contexts tablet-based interventions are most effective.

1.4. Purpose of the study

In this study, we attempted to evaluate the extant literature regarding evidence of generalization and maintenance effects against established criteria that account for the methodological rigor in SCED studies (i.e., IOA measures and the number of data points in generalization and maintenance). In addition, we conducted a meta-analysis to fill previously noted gaps in the literature by determining the following: (a) the extent to which tablet-based intervention studies for individuals with ASD collect generalization and maintenance data, (b) the type of strategies used to teach generalization and maintenance, and (c) the magnitudes of effect (i.e., Tau-U effect size) of tablet-based interventions with respect to generalization and maintenance effects. The specific research questions addressed in this study are as follows:

1. What is the status of the extant literature regarding evidence and methodological rigor of generalization and maintenance effects in SCED studies on tablet-based interventions for individuals with ASD?
2. What is the status of the extant literature regarding the collection of maintenance data, the collection of generalization data, and the type of generalization or maintenance teaching strategy used to promote sustained behavioral change?
3. What are the magnitudes of effect (i.e., Tau-U effect size) of tablet-based interventions with respect to generalization and maintenance of effects?

2. Method

The search procedures, screening, and exclusion based on quality of design mirrored ([*blinded for review]).*

2.1. Search procedure

A systematic literature search was performed using *ERIC, Academic Search Complete, Professional Developmental Collection, Social Science Full Text,* and *PsycINFO*. The selection was limited to articles appearing in the English language and in peer-reviewed journals but not restricted to publication year. The search query included the following terms: *autis*, ASD, pervasive developmental disorder, PDD,* and *Asperger*,* were combined with the terms *iPad, iOS, table-based computer, Proloque2Go,* and *iPod**. The search yielded 755, and once duplicates and articles without an author were removed, a total of 454 articles remained. The abstracts of the resulting articles were screened to exclude if the article did not include individuals with ASD, resulting in a total of 103 articles for further analysis.

2.2. Selection procedure

To be included in this review, an article had to meet the following inclusion criteria. First, the study had to include at least one participant who had a diagnosis of ASD. Second, the study had to use a tablet-based computer as an intervention component. Third, the study had to utilize a single-case research methodology and presented participant results in line graph format. Fourth, the study had to be published in English and in a peer-reviewed journal. As a result of the screening, 68 articles were identified to meet the initial inclusion and exclusion criteria. To evaluate design quality of these 68 articles, all identified articles were then evaluated based on the researched developed the design standards adapted from the What Works Clearinghouse (WWC; Kratochwill et al., 2010/2014). These standards evaluated whether (1) the independent variable was systematically manipulated, (2) the inter-observer agreement (IOA) data were collected, (3) the IOA data were reported on at least 20% of the data points in each phase/condition, (4) the resulting IOA data were 80% or higher or at least 0.60 measured by *Cohen's kappa* coefficient, (5) at least three replications of intervention effects were demonstrated through the design, and (6) a minimum number of data points was collected per phase. As a result of the design standards evaluation, a total of 36 articles met the preset design standards and 32 were excluded from further analysis.

Following the application of design standards, an ancestral search and hand search were conducted to increase the likelihood that all the potentially relevant studies were identified. For the ancestral search, the reference lists of studies that met the inclusion criteria were reviewed, resulting in a total of 12 articles. Among the 12 articles, none of them were identified to meet the design standards, and therefore, included in further analysis. The hand search covering journals published from January 2010 to May 2017 was conducted for the journals containing the articles meeting the design standards. This hand search yielded five articles, and three of them met the design standards. As a result of the additional screenings, a total of 39 articles were included in this review for further analysis when in addition to the 36 articles screened prior to the additional searches.

2.3. Screening for maintenance and generalization data

The 39 included articles were evaluated to identify whether they collected data in maintenance and/or generalization sessions on dependent variables. Generalization was considered to have occurred if a therapeutic behavioral change was observed over novel behaviors, persons, and/or settings in the absence of specific teaching component. With regard to maintenance of effects, maintenance was considered to have occurred if a behavioral change was detected after completion of a targeted intervention. Of the 39 studies, 21 articles included measures on generalization and/or maintenance of effects. Across the 21 articles that reported generalization and/or maintenance of effects, seven articles collected both generalization and maintenance data. A total of 14 articles reported either generalization or maintenance data, which is five included generalization data and nine reported maintenance data.

2.4. Application of design standards: evidence of generalization and maintenance effects and methodological rigor

Although there is no consensus on what quality indicator should be used over the others, most researchers have reached general agreement on several standards that must be reported in a single-case study for evaluating treatment effect and to be considered of high quality (Horner et al., 2005; Kratochwill, 2013). To date, the standards developed by WWC (Kratochwill et al., 2010/2014) are most frequently cited and include the following quality indicators: (a) systematic manipulation of the independent variable, (b) repeated measurements of the outcome variables by more than one evaluator, (c) three attempts to demonstrate an treatment effect, and (d) at least three data points collected in each condition/phase. Among these, we adopted two indicators (i.e., evaluation of IOA and the number of data points collected in each condition/phase) that were applicable to assess maintenance and generalization of treatment effects.

The maintenance and generalization data that met the basic design criteria were evaluated using a researcher designed rubric developed by Neely et al. (2017). The rubric was structured on the basic design standards established by WWC (Kratochwill et al., 2010/2014) and included four of the standards. For the first Design Standard, a score of 1 was assigned if IOA data on outcome variables were collected in generalization and/or maintenance and a score of 0 was assigned if IOA data were not reported. For the second Design Standard, a score of 1 was given if IOA was collected at least 20% of data points within each generalization and/or

maintenance phase and a score of 0 was given if it was not met. For the third Design Standard, a score of 1 was assigned if the resulting IOA met minimum thresholds, defined as 0.80 for percentage agreement indices or 0.60 for *kappa* measures and a score of 0 was assigned if it was not met. For the fourth Design Standard, a score of 1 was given if the generalization and/or maintenance phases included three or more data points and a score of 0 was assigned if the phases included one or two data points.

2.5. Coding of descriptive study characteristics

To extract descriptive data regarding the measures of generalization and maintenance data, the categories were coded based on the coding schema developed by (Neely et al., 2016). The studies included in this review were summarized based on the following variables: (a) generalization dimension, (b) generalization assessment design, (c) maintenance assessment design, (d) maintenance and generalization teaching strategy, and (e) latency to maintenance probes.

The generalization dimension included three categories: (a) setting (i.e., the data were collected across settings), (b) material or behavior (i.e., the data were collected across materials or behaviors, and (c) person (i.e., the data were collected across persons). The generalization assessment design contained four variables: (a) single probe (i.e., only one probe was collected in a post-intervention session), (b) multiple probes (i.e., more than one probe were collected in the duration of the study), and (c) continuous probes (i.e., the data were collected during the baseline, intervention, and post-intervention session). Within the multiple probes variable, studies that collected generalization data both in a pre- and post-intervention phase were subcategorized as pre-post multiple probes.

The maintenance assessment design variable included three categories: (a) single probe (i.e., only one probe was collected in a post-intervention phase), (b) multiple probes (i.e., more than one probe were collected in the duration of the study), and (c) sequential withdrawal (i.e., intervention components were sequentially withdrawn in consecutive experimental phases). In addition, description of the latency to collection of the maintenance data was coded (i.e., 4-week follow-up).

Eight categories were coded for the maintenance and generalization teaching strategy variables. First, if no specific generalization and/or maintenance teaching strategy was identified, the study was coded as train and hope (e.g., Smith, Ayres, Mechling, & Smith, 2013; van der Meer, Beamish, Milford, & Lang, 2015). Second, if the participant had to meet a predetermined criterion before collecting generalization and/or maintenance data, the study was coded as train to criterion and hope (e.g., Alzrayer, Banda, & Koul, 2017; Cihak, Fahrenkrog, Ayres, & Smith, 2009; Drysdale, Lee, Anderson, & Moore, 2015; Jowett, Moore, & Anderson, 2012; Kellems and Morningstar, 2012; Lorah et al., 2013). Third, if the instruction was delivered in a setting in which the participant was expected to present the targeted responses, the study was coded as train in natural setting (e.g., Mancil, Lorah, & Whitby, 2016). Fourth, if a study transferred behavioral contingencies from a contrived reinforcer to a natural reinforcer, this was coded as introduce to natural maintaining contingencies. Fifth, if multiple examples were used to train the participant prior to collecting generalization and/or maintenance data, the study was coded as multiple exemplars. Sixth, if the stimuli from the natural setting were used in the training setting, this was coded as program common stimuli (e.g., Alexander, Ayres, Smith, Shepley, & Mataras, 2013; Bouck, Savage, Meyer, Taber-Doughty, & Hunley, 2014; Kagohara et al., 2010; Nepo, Tincani, Axelrod, & Meszaros, 2015; Siegel & Lien, 2014; Spriggs, Knight, & Sherrow, 2015; Strasberger and Ferreri, 2014). Seventh, if a study utilized two or more teaching strategies, this was coded as multiple strategies. Eighth, if a study did not fit to any of the categories, this was coded as others.

2.6. Data extraction and analysis

In the current analysis, tau for nonoverlap with baseline trend control (Tau-U; Parker, Vannest, Davis, & Sauber, 2011) was calculated. Tau-U has several advantages over other nonoverlap indices. First, as opposed to most indices including percentage of nonoverlapping data (PND), improvement rate difference (IRD), and NAP, Tau-U can control undesirable baseline trend (Parker, Vannest, Davis et al., 2011). Second, Tau-U computes all data points in baseline and intervention phases so outliers are less likely to affect the results (Parker, Vannest, & Davis, 2011). Third, Tau-U shows the robustness to autocorrelated data (Parker, Vannest, Davis, 2011). However, it should be noted that Tau-U has several limitations (Parker, Vannest, Davis, 2011; Tarlow, 2017). First, baseline trend correction often increases the probability of Type 1 error (or false positive results). Second, Tau-U values are often inflated and not bound between -1 and $+1$. Third, Tau-U shows insensitivity to the magnitude of change between baseline and intervention phases. Fourth, although Tau-U performs reasonably well with autocorrelation, a certain level of serial dependency (i.e., > 0.20) exists in most SCED studies and this can distort Tau-U values. Despite its weaknesses, given the above-mentioned strengths of Tau-U, Tau-U was used and interpreted as an effect size (i.e., small effect = 0 to 0.65, moderate effect = 0.66 to 0.92, large effect = 0.93 to 1.00; Parker, Vannest, Davis et al., 2011). For the purpose of this study, omnibus Tau-U effect sizes were obtained. The following procedures were performed to calculate omnibus effect sizes for each moderator variable: (1) the individual Tau-U scores were weighted by the inverse of its variance, (2) the weighted values were added together, and (3) the sum was divided by the sum of weights (see Parker & Vannest, 2012).

Similar to other nonoverlap metrics, Tau-U accounts for data overlap in addition to trend control in baseline. This makes it easier for users to compute Tau-U since data can be extracted by hand using a rank order approach (Parker, Vannest, Davis, 2011). To get a rank order for data points in a graph, a horizontal line was drawn through each data point. A data point plotted at the lowest level on the graph was ranked 1, and a data point plotted on the second lowest was ranked 2. If two or more data points were plotted at the same level, same rank was given to the data. This procedure was repeated until all the data points in the graph were ranked. In the case that the study was designed to reduce unwanted behaviors and decreases in the behaviors indicated an improvement, rankings were reversed from top to bottom (i.e., a data point plotted at the highest level on the graph was ranked 1). The data in the generalization and maintenance conditions were contrasted with both the baseline and intervention conditions within a participant.

Only two phases were contrasted at a time (i.e., baseline vs generalization and intervention vs generalization, baseline vs maintenance and intervention vs maintenance). Further, to preserve the independence of all contrasts, each phase contrast in the design was only evaluated once. If a study utilized an alternating treatments design with no-treatment control condition, data were not extracted from the study (i.e., Mancil et al., 2016).

2.7. Inter-rater reliability

In this review, all inter-rater reliability (IRR) scores were calculated using point-to-point agreement. For the initial inclusion and exclusion review, a second rater replicated the literature search using the same parameters as the first rater and conducted the initial screening for 264 of the 454 articles (58%). The IRR for the inclusion and exclusion review was 94%. For accurate evaluation of the studies based on application of design standards, an independent rater rated 75% of the 68 studies (n = 51), resulting in 92% of IRR. To ensure accurate coding of the studies by moderator variables, a second independent reviewer rated 76% (n = 30) of the studies included in this review, yielding 92% of IRR. A second rater coded 100% (n = 19) of articles for evaluation of evidence of generalization and maintenance effects based on a researcher designed rubric structured on the basic design standards, resulting in 100% of IRR. For accurate raw data extraction, a second reviewer extracted data for 13 of the 20 studies (61%). The resulting IRR was 98% (range, 92.45–100%). Any disagreement was resolved by discussion until consensus was reached.

3. Results

Of the 39 articles evaluated against the preset inclusion and exclusion criteria, a total of 21 studies collected measures regarding the generalization and/or maintenance of the dependent variables. Across the 21 articles, 12 articles assessed for generalization of effects, and 16 articles assessed for maintenance of effects. Additionally, a total of 15 articles reported IOA, 14 of which collected IOA on at least 20% of data points and met minimum quality IOA thresholds.

3.1. Evidence of generalization and maintenance effects and methodological rigor

Table 1 shows the results of evaluation on evidence of generalization and maintenance effects. Overall, of the 12 articles that assessed for generalization of effects, three studies met all the design standards evaluated in this review. Nine studies reported that they collected an IOA on outcome variables in the generalization phases. All nine studies also collected IOA on at least 20% of data

Table 1
Evidence of generalization and maintenance effects.

Study	Generalization Evaluation				Maintenance Evaluation			
	Interobserver Agreement			Data Points	Interobserver Agreement			Data Points
	DS#1	DS#2	DS#3	DS#4	DS#1	DS#2	DS#3	DS#4
Aldi et al. (2016)	NA	NA	NA	NA	1	1	1	0
Alexander et al. (2013)	1	1	1	0	1	1	1	0
Alzrayer et al. (2017)	0	NA	NA	0 (different across phases)	NA	NA	NA	NA
Bereznak, Ayres, Mechling, and Alexander (2012)	NA	NA	NA	NA	0	NA	NA	0 (different across phases)
Bouck et al. (2014)	NA	NA	NA	NA	0	NA	NA	0
Burckley et al. (2015)	1	1	1	0	1	1	1	0
Cihak et al. (2009)	NA	NA	NA	NA	1	1	1	0
Drysdale et al. (2015)	0	NA	NA	1	1	1	1	1
Jowett et al. (2012)	1	1	1	0	1	1	1	0
Kagohara et al. (2010)	NA	NA	NA	NA	1	1	1	1
Kellems and Morningstar (2012)	NA	NA	NA	NA	1	1	1	0
Lorah et al. (2013)	NA	NA	NA	NA	0	NA	NA	0
Mancil et al. (2016)	1	1	1	1	1	1	1	1
Nepo et al. (2015)	1	1	1	0 (different across phases)	NA	NA	NA	NA
Siegel and Lien (2014)	1	1	1	1	NA	NA	NA	NA
Smith et al. (2013)	NA	NA	NA	NA	0	NA	NA	0
Spooner, Ahlgrim-Delzell, Kemp-Inman, and Wood (2014)	1	1	1	1	1	1	1	0 (different across phases)
Spriggs et al. (2015)	1	1	1	1	NA	NA	NA	NA
Strasberger and Ferreri,2014	1	1	1	0	1	1	1	0
van der Meer et al. (2015)	NA	NA	NA	NA	1	NA	NA	0
Weng and Bouck (2014)	0	NA	NA	0	NA	NA	NA	NA

Note. DS#1 = inter-observer agreement was reported; DS#2 = inter-observer agreement was collected at least 20% of data points within each generalization and/or maintenance phase; DS#3 = inter-observer agreement met minimum thresholds; DS#4 = number of data points per generalization and/or maintenance phase.

Table 2
Studies descriptive.

Study	Generalization Dimension	Generalization Assessment Design	Maintenance Assessment design	Latency to Maintenance Probe	Generalization/Maintenance Teaching Strategy
Aldi et al. (2016)	Not collected	Not collected	Single probe and multiple probes	3–4 weeks	Multiple strategies: Train to criterion and hope/Train in natural setting
Alexander et al. (2013)	Setting	Multiple probes: Pre-post multiple probes	Multiple probes	1.5–6 weeks	Programming common stimuli
Alzrayer et al. (2017)	Behaviors and materials	Multiple probes (post-intervention)	Not collected	Not collected	Train to criterion and hope
Bereznak et al. (2012)	Not collected	Not collected	Single probe and multiple probes	Not reported	Train to criterion and hope (maintenance data points without the return of the phone)
Bouck et al. (2014)	Not collected	Not collected	Multiple probes	4 weeks	Programming common stimuli
Burckley et al. (2015)	Behaviors and materials	Single probe (post-intervention)	Single probe	Not reported	Generalization: Programming common stimuli/ Maintenance: Train and hope
Cihak et al. (2009)	Not collected	Not collected	Multiple probes	9 weeks	Train to criterion and hope
Drysdale et al. (2015)	Setting	Continuous probes	Multiple probes	4 weeks	Train to criterion and hope
Jowett et al. (2012)	Setting/ Behaviors and materials	Multiple probes	Multiple probes	6 weeks	Train to criterion and hope
Kagohara et al. (2010)	Not collected	Not collected	Multiple probes	10 weeks	Programming common stimuli
Kellems and Morningstar (2012)	Not collected	Not collected	Single probe and multiple probes	Not reported	Train to criterion and hope
Lorah et al. (2013)	Not collected	Not collected	Single probe and multiple probes	Not reported	Train to criterion and hope
Mancil et al. (2016)	Setting/Persons	Multiple probes (post-intervention)	Multiple probes	2 weeks	Train in natural setting
Nepo et al. (2015)	Setting/ Behaviors and materials	Multiple probes (post-intervention)	Not collected	Not collected	Programming common stimuli
Siegel and Lien (2014)	Setting/ Behaviors and materials	Multiple probes (post-intervention)	Not collected	Not collected	Programming common stimuli
Smith et al. (2013)	Not collected	Not collected	Single probe	Not reported	Train and hope
Spooner et al. (2014)	Behaviors and materials	Multiple probes (post-intervention)	Multiple probes	Not reported	Multiple strategies: Programming common stimuli/ Train to criterion and hope
Spriggs et al. (2015)	Behaviors and materials	Multiple probes: Pre-post multiple probes	Not collected	Not collected	Programming common stimuli
Strasberger and Ferreri, 2014	Setting	Single probe (post-intervention)	Multiple probes	Not reported	Programming common stimuli
van der Meer et al. (2015)	Not collected	Not collected	Multiple probes	Not reported	Train and hope
Weng and Bouck (2014)	Setting	Single probe and multiple probes	Not collected	Not collected	Multiple strategies: Program common stimuli/Train to criterion and hope

points within each generalization phase and met the minimum quality thresholds of IOA (i.e., 0.80 for percentage agreement indices or 0.60 for *kappa* measures). Considering the number of data points per phase, five studies included more than three data points in each generalization phase while five studies collected one or two data points in each phase. Two studies included one or two data points in some generalization phases and more than three data points in other phases.

Overall, of the 16 articles that assessed for maintenance of effects, two studies met all the design standards. A total of 12 studies reported that they collected an IOA on outcome measures in the maintenance phases. All the 12 studies also collected IOA on at least 20% of data points within each generalization phase and met the minimum quality thresholds of IOA (i.e., 0.80 for percentage agreement indices or 0.60 for *kappa* measures). Considering the number of data points per phase, three studies collected more than three data points per maintenance phase. Across the 16 studies, a total of 11 studies included one or two data points per maintenance phase. Two studies included one or two data points in some generalization phases and more than three data points in other phases.

3.2. Generalization dimension

Table 2 presents a summary of each study according to the evidence of effects, generalization dimension, generalization assessment design, maintenance assessment design, maintenance and generalization teaching strategy, and latency to maintenance probe. Of the 12 articles that assessed for generalization of effects, four articles assessed generalization of effects across settings, and four articles evaluated generalization of effects across behaviors and materials. Four articles assessed generalization of effects by combining of more than one dimension.

3.3. Generalization assessment design

Of the 12 studies that collected measures regarding the generalization of effects, two utilized a single probe design in a post-intervention condition and one utilized a continuous probes design. A total of eight studies utilized a multiple probes design, and of the eight studies, two collected generalization data both before and after the intervention (i.e., a pre-post multiple probes design). One study utilized a single probe design for some participants and a multiple probes design for others.

3.4. Maintenance assessment design

Of the 16 studies that collected measures regarding the maintenance of effects, ten studies collected multiple data points and two conducted a single probe design. Four studies conducted a single probe design for some participants and multiple probes design for others.

3.5. Latency to maintenance probe

Across the 16 articles, only a half of the studies (n = 8) reported the latency to collection of the maintenance data. Data was collected anywhere between 1.5 weeks and 10 weeks after the final session of the intervention.

Table 3
Number of studies, participants, analyses, and Tau-U effect sizes: generalization/maintenance teaching strategy.

	Number of studies	Number of Participants	Number of Analyses	Group Tau-U [CI_{90}]
Baseline vs Generalization				
Train to Criterion and Hope	3	5	17	0.84 CI_{90} [0.67, 1.00]
Multiple Strategies	2	7	7	0.76 CI_{90} [0.37, 1.00]
Programming Common Stimuli	6	20	31	0.83 CI_{90} [0.70, 0.97]
Intervention vs Generalization				
Train to Criterion and Hope	3	5	17	0.16 CI_{90} [0.00, 0.31]
Multiple Strategies	2	7	7	0.58 CI_{90} [0.32, 0.84]
Programming Common Stimuli	6	20	31	0.16 CI_{90} [0.04, 0.28]
Baseline vs Maintenance				
Train to Criterion and Hope	6	18	38	0.96 CI_{90} [0.84, 1.00]
Train and Hope	3	5	9	0.86 CI_{90} [0.52, 1.00]
Multiple Strategies	2	6	14	0.96 CI_{90} [0.72, 1.00]
Programming Common Stimuli	4	11	14	0.91 CI_{90} [0.68, 1.00]
Intervention vs Maintenance				
Train to Criterion and Hope	6	18	38	0.13 CI_{90} [.03, 0.24]
Train and Hope	3	5	9	0.27 CI_{90} [−0.06, 0.60]
Multiple Strategies	2	6	14	0.47 CI_{90} [0.26, 0.68]
Programming Common Stimuli	4	11	14	−0.04 CI_{90} [−0.23, 0.15]

3.6. Generalization and maintenance teaching strategy

Across the 21 articles, four different teaching strategies were employed. A total of seven studies employed a programming common stimuli strategy, two utilized a train and hope strategy, and seven utilized a train to criterion and hope strategy. Three studies employed multiple teaching strategies including a programming common stimuli, train to criterion and hope strategy, and train in natural setting. Burckley, Tincani, and Guld Fisher, 2015) employed a programming common stimuli as a generalization teaching strategy and a train and hope as a maintenance teaching strategy. In addition, Mancil et al. (2016) utilized a train in natural setting strategy.

3.7. Effect size analysis

Data from this study yielded a total of 130 separate AB contrasts (i.e., baseline/intervention vs. generalization/maintenance) from 20 articles with 56 participants. The omnibus Tau-U effect size for baseline to generalization comparisons was 0.85 CI_{90} [0.76, 0.95], indicating a moderate effect. For the contrasts between intervention and generalization comparisons, the omnibus Tau-U effect size was 0.18 CI_{95} [0.09, 0.27], indicating a small effect. The omnibus Tau-U effect size for baseline and maintenance comparisons was 0.94 CI_{90} [0.85, 1.00], indicating a large effect. For the contrasts between intervention to maintenance comparisons, the omnibus Tau-U effect size was 0.16 CI_{90} [0.08, 0.25], indicating a small effect.

For the analysis of the generalization teaching strategy, three variables were included (see Table 3). For baseline to generalization comparisons, two of the variables demonstrated moderate effects, ranging from 0.83 CI_{90} [0.70, 0.97] for a programming common stimuli strategy to 0.84 CI_{90} [0.67, 1.00] for a train to criterion and hope strategy. A moderate effect (Tau-U = 0.76 CI_{90} [0.37, 1.00]) was identified for multiple strategies. For intervention to generalization comparisons, two of the variables demonstrated small effects, ranging from 0.16 CI_{90} [0.04, 0.28] for a programming common stimuli strategy to 0.16 CI_{90} [0.00, 0.31] for a train to criterion and hope strategy. A small effect (Tau-U = 0.58 CI_{95} [0.32, 0.84]) was yielded for multiple strategies.

For the analysis of the maintenance teaching strategy, four variables were evaluated (see Table 3). For baseline to maintenance comparisons, all of the variables ranged from a moderate effect of 0.86 CI_{90} [0.52, 1.00] for a train and hope strategy to a large effect of 0.96 CI_{90} [0.84, 1.00] for a train to criterion and hope strategy. For intervention to maintenance comparisons, two of the variables demonstrated small effects, ranging from −0.04 CI_{90} [-0.23, 0.15] for a programming common stimuli strategy to 0.13 CI_{90} [0.03, 0.24] for a train to criterion and hope strategy. For a train and hope strategy, a small effect (Tau-U = 0.27 CI_{90} [−0.06, 0.60]) was yielded. Multiple strategies also yielded a small effect (Tau-U = 0.47 CI_{90} [0.26, 0.68]).

4. Discussion

This review and *meta*-analysis synthesized and analyzed 21 studies regarding the maintenance and generalization of tablet-based interventions for individuals with ASD. The included 21 studies were first summarized according to the evidence of effects, generalization dimension, generalization assessment design, maintenance assessment design, teaching strategy, and latency to maintenance probes. Raw data was then extracted and Tau-U effect sizes were calculated to determine the effectiveness of the generalization and maintenance strategies.

4.1. Evidence of generalization and maintenance effects and methodological rigor

The first purpose of this review was to evaluate the current status of the extant literature regarding evidence of generalization and maintenance effects against the existing criteria that account for the methodological rigor in SCED studies. To our knowledge, no previous studies have attempted to evaluate evidence of generalization and maintenance effects of tablet-based interventions in ASD-focused SCED studies regarding the methodological rigor. Overall, more than half of the studies included in this review collected IOA for at least 20% of sessions across each generalization and/or maintenance phase. This finding is consistent with those reported by Neely et al. (2015) indicating continuing improvement in the quality of IOA measures in autism-focused single-case research. Additionally, a majority of the studies met minimum quality thresholds for the IOA data reported, providing some evidence that these studies were internally valid. With regard to replication of treatment effects, the number of data points collected in each generalization and maintenance phase was also assessed. As a result, less than one third of the studies included more than three data points per phase, which reduces the methodological rigor of the studies in terms of evidence of generalization and maintenance of effects of tablet-based interventions. However, it should be noted that the quality indicator adopted in the current review in order to assess replication of treatment effects (i.e., the number of data points per phase/condition) was developed with the intention of evaluating baseline and intervention phases (Kratochwill et al., 2010/2013). This leads to the question of whether the WWC standards for phase length can be meaningfully applied to generalization and maintenance. Furthermore, this review only included the studies that either met the basic design standards or met them with reservations. Therefore, considering the application of the WWC standards for generalization and maintenance of effects and the number of studies excluded from this review in the basic design standards evaluation stage (N = 46), these results should be interpreted with caution.

4.2. Descriptive review

The second purpose of this review was to synthesize the literature regarding maintenance and generalization of tablet-based

interventions. Of the 39 studies that met design criteria, 21 collected maintenance or generalization data. The results of this review support the findings of previous reviews and meta-analysis identifying acquisition of skills, rather than sustainability of skills, as the focus for autism research (e.g., Neely et al., 2015). This is alarming considering the popularity of these types of interventions. While the 5-3-20 rule from the WWC does not require generalization and maintenance data be collected for an evidence-based practice determination (Kratochwill, 2013), other rubrics place more emphasis on these measures (e.g., Reichow et al., 2008). Because of the importance of generalization and maintenance to the sustained impacts of these interventions, we cannot conclusively support tablet-based interventions without these data. In addition, given the results of previous reviews identifying the positive impact of tablet-based interventions for individuals with ASD, a focus on maintenance and generalization of effects is warranted (Alzrayer, Banda, & Koul, 2014).

A strength of the body of literature reviewed is the majority of studies collected multiple data points when they did collect maintenance and generalization data. Interpretation of single-case research data depends on the trend and slope of a data path (Kratochwill, 2013). A minimum of three data points is necessary to meet basic design standards, with more data points leading to stronger conclusions regarding the data set. While only a few studies collected three or more data points (e.g., Jowett et al., 2012), the fact that a majority of the maintenance and generalization collected at least two data points is a strength of the literature base.

Another strength of the literature base is the large percentage of studies that collected data regarding maintenance of intervention effects. Unfortunately, a majority of the studies either did not report the latency to the maintenance probes, or collected maintenance data less than 10 weeks following the conclusion of the intervention. This relatively short maintenance phase is consistent with other reviews and meta-analyses which identify maintenance data is typically collected within 6-months following the teaching phase (e.g., Neely et al., 2016; Palmen et al., 2012). Given these short follow-up times, the long term sustainability of these interventions following training phases is questionable.

While a majority of studies collected maintenance data, a smaller percentage collected generalization data. In addition, only two of the studies collected generalization baseline data (i.e., before the intervention). This potentially corresponds with a train and hope model identifying that maintenance and generalization of skills is typically secondary to acquisition of skills. It also brings into question the conclusiveness of the generalization data without a baseline generalization probe for comparison. Future research might consider not only collecting multiple maintenance and generalization data points but also to collect baseline generalization probes.

4.3. Effect size analysis

The second purpose of this review was to conduct effect size analysis to identify if there are any differential effects on generalization and or maintaining targeted skills dependent on instructional strategy. There were three generalization strategies employed: programming common stimuli, train to criterion and hope, and multiple strategies. There were small to moderate effects noted for all of the strategies when comparing baseline to generalization data. Large effects were noted for train to criterion and hope and programming common stimuli. This suggests that both of the strategies might facilitate larger generalization of effects for tablet-based interventions. Small to moderate effects were noted for all teaching strategies when comparing intervention data to generalization data suggesting that effects continued to improve during the generalization phase.

There were four maintenance strategies employed across the included studies: train and hope, train to criterion and hope, programming common stimuli, and multiple strategies. All of the strategies demonstrated large effects when comparing baseline to maintenance phases suggesting skills maintained above baseline levels. However, when comparing intervention to maintenance, multiple strategies showed continued growth during the maintenance phase, with small improvements noted for train and hope, train to criterion and hope, and programming common stimuli. Unfortunately, these results are limiting as the differential impacts of one strategy versus another cannot be identified within the multiple strategies category.

4.4. Conclusions, limitations, and implications for practice

This study is the first attempt to evaluate the extant literature regarding evidence of generalization and maintenance effects against the existing criteria that account for the methodological rigor in SCED studies. In addition, the data supports the use of a train to criterion and hope teaching strategy to promote generalization and maintenance of skills. While the data indicated that multiple strategies was also effective, it cannot be promoted as a promising strategy because it is unknown which of the strategies within this category was effective. There are a number of limitations and implications for future research that should be noted. First, the number of studies that included maintenance or generalization data was limited. Therefore, our analyses should be viewed with caution. However, these results do highlight a need for future researchers to investigate interventions designed to promote sustainability of skills beyond the training phase. Second, the four design standards evaluated in this review mainly accounted for the internal validity of the studies. Meanwhile, less attention was paid to generality of treatment effects relevant to external validity, such as how well each study described the baseline condition, independent variable, dependent variable, participants, and setting (e.g., Hitchcock et al., 2015; Horner et al., 2005; Maggin et al., 2013; Reichow et al., 2008; Wolery, 2013); therefore, we were unable to draw conclusions about the methodological rigor of these criteria. Third, although most of the studies did collect multiple maintenance or generalization data points, there is still a need to improve the rigor of this data collection. In particular, single-case researchers should aim to collect a minimum of three maintenance data points and a minimum of three generalization data points in both the baseline and generalization phase. Fourth, the possibility that publication bias exists in this data set cannot be ruled out due to the inclusion of only peer-reviewed research (Shadish, Zelinsky, Vevea, & Kratochwill, 2016). Lastly, the results of the effect size analysis should be interpreted with caution. Although Parker, Vannest, Davis et al. (2011) provides different percentile values of the Tau-U effect size

derived from their field test, the benchmarks are intended to be used for comparisons of adjacent phases (i.e., a baseline phase vs. an actually adjacent intervention phase). Therefore, it should be noted that the Tau-U effect size benchmarks might not be appropriate to apply when contrasting non-adjacent phases as performed in the current review.

In light of these limitations, there are a number of suggestions for practitioners regarding the use of tablet-based interventions for individuals with ASD. First, in the current analysis, studies were included only if they met the pre-set design standards or met them with reservations. As a result of the analysis, only three studies that collected generalization of effects (25%) were found to have met all the design standards. With regard to maintenance of effects, two studies (12.5%) met all the standards. Such findings indicate that SCED researchers are still in early stages of establishing the methodological rigor regarding generalization and maintenance of effects in SCED studies. This is likely due to difficultly identifying criteria that quantify evidence of generalization and maintenance effects. Therefore, it is suggested to establish a system for appraising generalization and maintenance procedures in SCED studies. When developing this system, researchers might consider including criteria to determine whether participants made socially relevant improvement, as assessed by social validity measures, in addition to measuring visual and statistical improvements in the data. A second recommendation is for research synthesis to collect data on the latency to maintenance measure when conducting systematic reviews and meta-analyses and report on these findings; this is particularly important given the recent efforts in the SCED literature to improve reporting (Tate et al., 2016).

A final suggestion is to plan for the maintenance and generalization of learned skills from the onset of the intervention. Stokes and Baer (1977) suggested a number of teaching strategies and this review highlights the utility of the train to criterion and hope and programming common stimuli methods. However, due to the limited data supporting these conclusions, practitioners are urged to evaluate their programs through continuous progress monitoring. In addition, practitioners should continue to provide ongoing support to ensure treatment fidelity and adherence to the intervention following the teaching phase. Overall, the evidence supporting tablet-based interventions is growing. However, there is a need to enhance the rigor of the literature base and further the evidence in regards to the maintenance and generalization of effects.

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The authors report no conflicts of interest. The authors alone are responsible for the content and writing of this article.

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